WORKED SOLUTION

STEP III 2001, Question 6

Pure • Vectors and Triangle Centres

STEP III 2001, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q6

The key idea

Everything is controlled by the normal vector $\mathbf{n} = \left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right)$ of the plane: a point $\mathbf{r}$ lies in the plane exactly when $\mathbf{r} \cdot \mathbf{n} = 1$, and the perpendiculars from O and M both run along $\mathbf{n}$. The orthocentre and circumcentre come out of two pieces of 3D geometry: OA, OB, OC are mutually perpendicular, and M is the centre of the box with edges OA, OB, OC, so it is equidistant from O, A, B and C. For the last part, the two perpendiculars are parallel, so a pair of similar triangles (or one line of vector algebra) does the rest.

Solution

Throughout, a, b, c are non-zero (otherwise the plane doesn't meet all three axes), so

$$A = (a, 0, 0), \qquad B = (0, b, 0), \qquad C = (0, 0, c),$$

and we write $S = \dfrac{1}{a^2} + \dfrac{1}{b^2} + \dfrac{1}{c^2} = |\mathbf{n}|^2 > 0$.

The centroid

Points of the line OM are $t\left(\frac{1}{2}a, \frac{1}{2}b, \frac{1}{2}c\right)$. Substituting into the plane's equation gives $\frac{1}{2}t + \frac{1}{2}t + \frac{1}{2}t = 1$, so $t = \frac{2}{3}$, and the meeting point is

$$G = \left(\tfrac{1}{3}a, \tfrac{1}{3}b, \tfrac{1}{3}c\right) = \tfrac{1}{3}(A + B + C),$$

which is the centroid of triangle ABC. Note for later that $\overrightarrow{OG} = \frac{2}{3}\overrightarrow{OM}$.

The perpendicular from O: the orthocentre

The perpendicular from O is the line $\lambda\mathbf{n}$. It meets the plane where $\lambda\,\mathbf{n}\cdot\mathbf{n} = 1$, that is $\lambda = 1/S$, at

$$H = \frac{1}{S}\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right).$$

We show that $AH \perp BC$. Using $\overrightarrow{BC} = (0, -b, c)$,

$$\overrightarrow{AH}\cdot\overrightarrow{BC} = \left(\frac{1}{aS} - a\right)\cdot 0 + \frac{1}{bS}\cdot(-b) + \frac{1}{cS}\cdot c = -\frac{1}{S} + \frac{1}{S} = 0.$$

Ask yourself Is there a reason this dot product had to vanish, rather than just happening to?

Yes. BC lies in the plane, so it is perpendicular to the normal OH; and BC lies in the yz-plane, so it is perpendicular to OA, which is along the x-axis. A line perpendicular to two non-parallel lines OA and OH is perpendicular to the whole plane OAH, and in particular to AH.

By the same argument with the letters permuted, $BH \perp CA$ and $CH \perp AB$. Since H lies in the plane of the triangle, $AH \perp BC$ says that H lies on the altitude from A, and similarly for the other two altitudes. So H is the orthocentre of triangle ABC.

The perpendicular from M: the circumcentre

The perpendicular from M is $M + \mu\mathbf{n}$. It meets the plane where $\frac{3}{2} + \mu S = 1$, so $\mu = -\dfrac{1}{2S}$, at

$$P = M - \frac{1}{2S}\,\mathbf{n}.$$

Now M is equidistant from all four of O, A, B, C: for example

$$MA^2 = \left(\tfrac{1}{2}a - a\right)^2 + \left(\tfrac{1}{2}b\right)^2 + \left(\tfrac{1}{2}c\right)^2 = \tfrac{1}{4}\left(a^2 + b^2 + c^2\right),$$

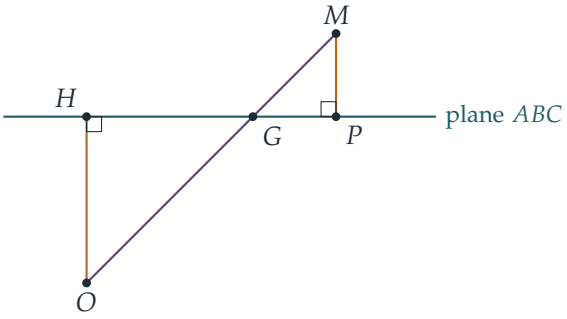
and MB^2, MC^2 (and MO^2) give the same value by symmetry. Since MP is perpendicular to the plane and PA lies in it, triangle MPA has a right angle at P, so by Pythagoras

$$PA^2 = MA^2 - MP^2,$$

and likewise $PB^2 = MB^2 - MP^2$ and $PC^2 = MC^2 - MP^2$. The right-hand sides are equal, so $PA = PB = PC$. The only point of the plane equidistant from the three vertices is the circumcentre, so P is the circumcentre of triangle ABC.

The final part

The perpendiculars OH and MP are both parallel to $\mathbf{n}$, so O, H, M, P lie in one plane, and the line OM crosses the triangle's plane at G. Also O and M are on opposite sides of the triangle's plane, since $O\cdot\mathbf{n} = 0 < 1$ and $M\cdot\mathbf{n} = \frac{3}{2} > 1$. Here is the cross-section through these four points:



Triangles OHG and MPG have right angles at H and P and equal vertically opposite angles at G, so they are similar, with ratio $OG : GM = 2 : 1$. Hence $HG : GP = 2 : 1$, and G lies on the segment HP.

The same conclusion drops out of the vectors directly. With $H = \frac{1}{S}\mathbf{n}$, $P = M - \frac{1}{2S}\mathbf{n}$ and $G = \frac{2}{3}M$,

$$\tfrac{1}{3}(H + 2P) = \tfrac{1}{3}\left(\tfrac{1}{S}\mathbf{n} + 2M - \tfrac{1}{S}\mathbf{n}\right) = \tfrac{2}{3}M = G.$$

So G lies on the segment from the orthocentre H to the circumcentre P and divides it in the ratio $HG : GP = 2 : 1$, as required. (If the triangle is equilateral then $H = P = G$ and there is nothing to prove.)

Ask yourself Does this prove the result for *every* triangle?

Not quite. For the argument to apply, a triangle has to be congruent to some triangle ABC of the type above. Its sides then satisfy $BC^2 = b^2 + c^2$, $CA^2 = c^2 + a^2$ and $AB^2 = a^2 + b^2$, so

$$AB^2 + AC^2 - BC^2 = 2a^2 > 0,$$

and similarly at B and C. By the cosine rule every angle is acute. Conversely, any acute triangle can be realised: solve these three equations for a^2, b^2, c^2, which are all positive. So the method proves the result for acute triangles. For a right-angled or obtuse triangle, here is a short planar argument that covers every case. Put the circumcentre at the origin, so the vertices $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ all have the same length. Let $\mathbf{h} = \mathbf{a} + \mathbf{b} + \mathbf{c}$. Then

$$(\mathbf{h} - \mathbf{a}) \cdot (\mathbf{b} - \mathbf{c}) = (\mathbf{b} + \mathbf{c}) \cdot (\mathbf{b} - \mathbf{c}) = |\mathbf{b}|^2 - |\mathbf{c}|^2 = 0,$$

so $\mathbf{h}$ is on the altitude from $\mathbf{a}$, and by symmetry on the other two altitudes: it is the orthocentre. The centroid is $\frac{1}{3}\mathbf{h} = \frac{1}{3}(\mathbf{h} + 2 \cdot \mathbf{0})$, which is the same $2 : 1$ statement.

Checks

Take $a = 1$, $b = c = 2$, so $S = \frac{3}{2}$. Then $H = \left(\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$, $P = \left(\frac{1}{6}, \frac{5}{6}, \frac{5}{6}\right)$ and $G = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$. Each satisfies $x + \frac{1}{2}y + \frac{1}{2}z = 1$. You can check that $\overrightarrow{AH} \cdot \overrightarrow{BC} = 0$ and $PA^2 = PB^2 = PC^2 = \frac{25}{12}$, and that $\frac{1}{3}(H + 2P) = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right) = G$. We also checked all of these facts in exact arithmetic for 3000 random choices of a, b, c of both signs.

Common slips

- Showing only $AH \perp BC$ and stopping. You need either all three altitudes, or a clear "by symmetry". You also need to say that H lies in the triangle's plane: a point off the plane can have $AH \perp BC$ without being on the altitude.
- Getting the ratio the wrong way round. The centroid is twice as far from the orthocentre as from the circumcentre.
- Claiming the "hence" covers every triangle. As shown above, it only reaches acute ones.

Beyond STEP

The line through the orthocentre, centroid and circumcentre is the *Euler line*. The fact $\mathbf{h} = \mathbf{a} + \mathbf{b} + \mathbf{c}$ (with the circumcentre at the origin) is the starting point for the nine-point circle, whose centre is also on the Euler line, halfway between H and the circumcentre. The trick of seeing a plane figure as a slice of a simple 3D one is a recurring idea in geometry. Coxeter's *Introduction to Geometry* and Coxeter and Greitzer's *Geometry Revisited* both treat the Euler line and the nine-point circle beautifully.

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